

Manasvin Surya

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EDUCATION

University of California, San Diego

M.S. Electrical and Computer Engineering (Machine Learning & Data Science)

Expected (March 2027)

United States

National Institute of Technology, Trichy

B.Tech. Electronics and Communication Engineering

August 2020 - May 2024

India

SKILLS

Languages: Python, C++, C, Java, SQL.

ML Systems & LLM Optimization: KV Cache Optimization (Prefix Reuse, LRU/TTL Eviction), vLLM, model serving, distributed inference, Speculative Decoding, FlashAttention, LoRA, PEFT.

Backend & Data Infrastructure: Spring Boot, Microservices, SQLAlchemy, MongoDB, Redis, RESTful APIs.

Frameworks & Platforms: PyTorch, LangChain, Docker, Kubernetes, Git, MLFlow, DVC, Spark.

Selected Courseworks: Hardware Acceleration & optimization for Deep Learning, Big Network Data, Agentic AI for Smart Grids, Data Structures & Algorithms, DBMS, Operating Systems.

EXPERIENCE

Standard Chartered GBS

Software Development Engineer

August 2024 - August 2025

India

- **Optimized query performance by 30%** and **accelerated credit processing by 15%** by migrating databases from Oracle database to PostgreSQL and refactoring schema to eliminate redundant joins.
- **Reduced KYC processing latency by 40%** by architecting a resilient extraction pipeline for OCR and NER models, significantly improving system throughput and downstream triage efficiency for compliance workflows.

Xu Labs, Carnegie Mellon University

Machine Learning Research Intern

January 2024 - March 2024

United States

- Architected **distributed multimodal pipelines** using **FlashAttention** and **sparse kernels** to scale Cryo-ET retrieval on **multi-node A100/H100 clusters**; reduced manual interventions by **20%** via **drift-aware reliability metrics** while achieving **ROUGE 0.782**.

Standard Chartered GBS

Software Development Intern

May 2023 - July 2023

India

- **Achieved F1 0.89 retrieval accuracy** for operational decision-making by architecting a **production-grade RAG infrastructure**, **optimizing the FAISS vector indexing pipeline** for low-latency document retrieval.
- Accelerated operational workflows at scale by engineering a high-throughput support system that replaced manual policy lookup with an **automated LLM-based query resolution engine**.

Samsung Research Institute

Machine Learning Intern

August 2022 - February 2023

India

- **Optimized edge-device inference to a 3% latency overhead** by implementing LoRA-based PEFT for open-domain QA, **ensuring 81.25% accuracy** on resource-constrained hardware.
- **Engineered a 20x reduction in trainable parameters** (using only 5% of weights) by architecting a modular LoRA adapter framework for BART, **maintaining 95% of full fine-tuning performance**.

PROJECTS

Cache-Aware LLM Inference: KV and Cross-Request Prefix Reuse

December 2025

- Achieved **2.80x higher throughput** and **63.3% latency reduction** by architecting a production-grade LLM inference server with KV cache optimization and **LRU+TTL eviction policies**.
- Optimized **per-token decode time by 66.7%** and reached **near-zero TTFT** by engineering a prefix-sharing mechanism that maximizes KV-cache hits under concurrent-request benchmarks.

LLM-powered semantic query engine - Standard Chartered GBS

March 2025

- **Increased query accuracy to 88.2%** and **execution success to 84.7%** by architecting a RAG-based schema grounding layer to replace standard few-shot prompting for customer credit data.
- Engineered a resilient SQL execution pipeline using SQLAlchemy for automated schema validation and query sanitization, enabling natural language access to complex relational databases.

Novel QA System Using Hierarchical NER System - AGS Healthcare, USA

January 2024

- **Achieved 0.89 precision and 0.767 MRR** by initiating a hierarchical NER tagging architecture using RoBERTa, enabling the system to identify and rank fine-grained symptom data.
- **Improved retrieval relevance by 22%** by engineering a multi-level ranking algorithm that prioritized symptoms, reducing the search space for QA resolution.

HACKATHONS

Winner - NSF HDR Scientific Modeling out of distribution: Neural Forecasting, UCSD

January 2026